

### ChE 313 – Transport phenomena III –Fall 2019

#### Student Learning Outcome 1

Rubric for assessment problem on the final exam (Problem #D)

Derive the equation to calculate the steady-state mass flux through a membrane.

| <b>SLO 1<br/>Performance<br/>Indicator (PI)</b>                                                                                                | <b>(1) Unsatisfactory</b>                                                                                                                        | <b>(2) Developing</b>                                                                                                                     | <b>(3) Satisfactory</b>                                                                                                                                       |
|------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Apply mathematical and scientific principles to formulate models and systems relevant to the subject<br>Problem D(a), 10 pts                | Able to successfully combine mathematical and/or scientific principles to formulate models and systems relevant to chemical engineering (<6 pts) | Chooses a mathematical model or scientific principle that applies to an engineering problem, but has trouble in model development (6 pts) | Does not understand the connection between mathematical models and the system or process to be analyzed or designed<br><br>(7-10 pts)                         |
| 2. Solve engineering problems by using the concepts of algebra, calculus, differential equations and/or linear algebra<br>Problem D(b), 10 pts | applies concepts of algebra, calculus, differential equations and/or linear algebra to solve chemical engineering problems (<6 pts)              | Shows nearly complete understanding of applications of calculus and/or linear algebra in problem-solving<br><br>(6 pts)                   | Does not understand the application of algebra, calculus, differential equations and/or linear algebra in solving chemical engineering problems<br>(7-10 pts) |

Averages:

Performance Indicators: (1) 2.35 (2) 2.69

Overall Average: 2.52

Average scores were fairly uniform although there was wide variation among students. Clearly, most students were strong in formulating the engineering system and model equations, and a majority 18/26 were satisfactory or higher in applying the engineering principles to solve the three problems. In the future, I plan to use multiple problems to assess one learning outcomes in the beginning, middle and end of the semester, so we can track the growth of the students more accurately.

**D. (20 pts)** A liquid containing dilute solute A at a concentration  $c_1 = 3 \times 10^{-2}$  kmole/m<sup>3</sup> is flowing rapidly past a membrane of thickness  $L = 3.0 \times 10^{-5}$  m. The distribution coefficient  $K = 1.5$  and  $D_{AB} = 7.0 \times 10^{-11}$  m<sup>2</sup>/s in the membrane. The solute diffuses through the membrane, and its concentration on the other side is  $c_2 = 0.5 \times 10^{-2}$  kmole/m<sup>3</sup>. The mass-transfer coefficient  $k_{c1}$  is large and can be considered as infinite, and  $k_{c2} = 2.0 \times 10^{-5}$  m/s.

(a) Derive the equation to calculate the steady state flux  $N_A$ .

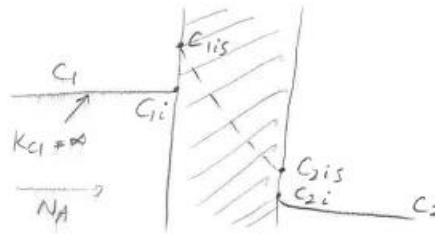
(b) Calculate the flux and the concentrations ( $c_{1iS}$ ,  $c_{2iS}$  and  $c_{2i}$ ) at the membrane interfaces.

# Solution

problem D

part a

on the left side  $\rightarrow C_1 = C_{1i}$



$$N_A = \frac{C_1 - C_2}{\frac{1}{K_{c1}} + \frac{1}{P_m} + \frac{1}{K_{c2}}} \xrightarrow{K_{c1} = \infty} N_A = \frac{C_1 - C_2}{\frac{1}{P_m} + \frac{1}{K_{c2}}}$$

part b

$$P_M = \frac{D_{AB} K'}{L} = \frac{7.0 \times 10^{-11} (1.5)}{3.0 \times 10^{-5}} = 3.5 \times 10^{-6} \text{ m/s}$$

$$N_A = \frac{C_1 - C_2}{\frac{1}{P_M} + \frac{1}{K_{c2}}} = \frac{3.0 \times 10^{-2} + 0.5 \times 10^{-2}}{\frac{1}{3.5 \times 10^{-6}} + \frac{1}{2.02 \times 10^{-5}}} = 7.458 \times 10^{-8} \frac{\text{kg mol}}{\text{s m}^2}$$

To calculate  $C_{2i}$ :

$$N_A = 7.458 \times 10^{-8} = K_{c2} (C_{2i} - C_2) = 2.02 \times 10^{-5} (C_{2i} - 0.5 \times 10^{-2})$$

$$\rightarrow C_{2i} = 0.869 \times 10^{-2} \frac{\text{kg mol}}{\text{m}^3}$$

$$K' = 1.5 = \frac{C_{2is}}{C_{2i}} = \frac{C_{2is}}{0.869 \times 10^{-2}} \rightarrow C_{2is} = 1.304 \times 10^{-2} \frac{\text{kg mol}}{\text{m}^3}$$